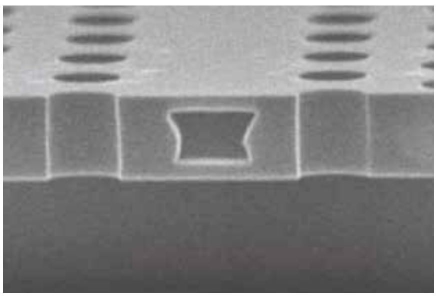
Data Limitations
and future routers

Limited bandwidth is one thing that is holding us back and at the rate at which mobile internet devices are being lapped up by consumers, the cap on bandwidth needs to be lifted. It is estimated that by 2016, the number of internet devices will outnumber the world's population. Blame it on smart phones and tablets, as the world lives half its life out online. We've been using the same routers for the past so many years to connect to the internet, which really doesn't do justice to the amount of data flowing through the future-proof fiber optics cables we have flowing underwater. And even though Wi-Fi removes this hurdle, with data flowing in its purest form, it still must respond to a transmitter or tower that is hard-wired to the vast global network of cables. Giving everyone their own personal fiber optics cable isn't practical or economical to overcome this bottleneck.

Telecom Regulatory Authority of India has proposed the National Broadband Plan (NBP) under which the Bharat Broadband Network Limited (BBNL) is supposed to manage and operate the National Optic fiber Network (NOFN). The main objective of the NOFN is to provide 2,50,000 villages with internet connectivity via fiber optic technology, to help out in many areas such as education, entertainment, environment, e-governance and other services. A total of 5,00,000 km of optic fiber cable is likely to be laid under the NOFN. While the idea seems tempting, at the end of the day it is a Government of India initiative, so naturally, the actual timeline should be taken with a pinch of salt.

While we still grapple with high-speed internet connectivity, Japanese telecom giant NTT is working on what could possibly be an important component in future routers. Researchers at NTT have build an optical random access memory (o-RAM) chip to make efficient storage buffers for internet routers. They have built a 4-bit prototype which operates on 40 Gbps whereas NTT is targetting 10-kilobit to 1-megabit memory chips for future optical routers. The cell in the o-RAM is actually a nanoscale photonic crystal - which is a material that channels light in small spaces. It operates between light-transmitting (1) and light-blocking states (0) to create digital signals. The technology is very light on energy consumption, using just 30 nanowatts of power and retaining data for one microsecond. This leads to more efficiency than normal electrical routers, and helps maintain the high data rate transmitted via fiber optics cables. Note that even though optic fiber cables have great bandwidth, the data is routed to us via routers which have electrical innards – thereby creating a bottleneck. So even though the data enters and leaves a router via beams of light, within the router the optical to electrical signal conversion causes the speeds to reduce.



NTT's optical RAM

Researchers at the University of Southampton in England have produced optical fibers which can transfer data at 99.7 per cent the speed of light. These optical fibers transfer data at the rate of 73.7 terabits per second which is much faster than the 40 gigabits per second that is done by fiber optic links today. The researchers employ a hollow optical fiber mostly made of air, as light travels around 31 per cent slower through silica glass (which makes the normal optical fiber). While such high data transfer may not make much sense to us anytime soon, as normal optic fiber cables are good enough, but for short stretches such as in data centres and supercomputer interconnects these high-speed fibers can be very significant.

* Holographic Data Storage

Hard drives have remained similar in design over the years. Of course capacities have been increasing and areal density is going up. But the principal mechanism of the hard drive still involves spinning discs. Off late, we have been seeing a lot of hybrid hard drives which employ some gigabytes of SLC or MLC NAND chips which speed up the access times and improves the overall performance of the drive. But how long will we rely on spinning hard drives? Fine, for personal use we have ways to backup data and there's always the cloud storage option. But deep archival storage requires storage over longer period of time such as decades. Magnetic media maxes out at five to ten years. Holographic storage stores data in three-dimensional volume as opposed to two dimensional data storage. It also employs write-once-read-many (WORM) archival storage technique. Holographic storage can in theory store about four gigabits per cubic millimeter – a huge number and it has a lifespan of over 50 years. Ten years down the line when digital media will explode, expect to see holographic storage for consumer use, but the first use-case will be for archiving purpose.